Total Marks: 15

EE2019 – Analog Systems and Lab

Duration: 65 min

Question-1

(10 marks)

Figure 1 shows a block diagram of an integrator. The integrator is realized using a real opamp whose small signal model is also given. Assume no voltage and current limits on node voltages and branch currents. The component values are as follows:

R_1 = 10kΩ, C_F = (1/2π)*1nF, C_P = (1/2π)*10pF, C_O = (1/2π)*100pF, r_o = 100kΩ, and gm = 10mA/V.

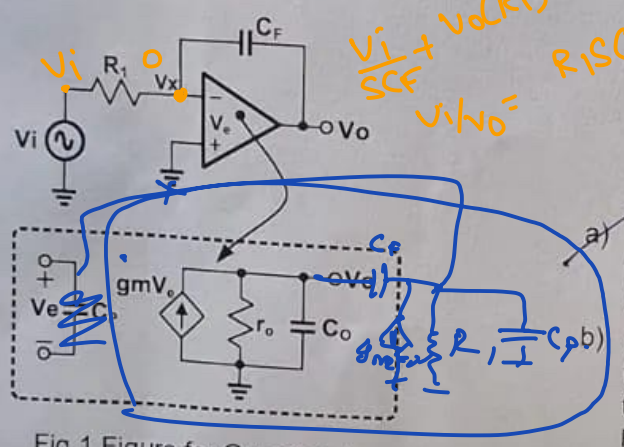


Fig.1 Figure for Question-1.

a) Find open loop gain (LG(s)) expression for the above loop. Draw Bode magnitude and phase plots of the loop gain. *(2 marks)*

b) Find the closed loop transfer function, H(s) = Vo(s)/Vi(s) in terms of loop parameters. *(2 marks)*

c) Find poles and zeros for H(s) in 1(b). Draw Bode magnitude and phase plots for H(s). *(2 marks)*

d) Find the frequency range (lower and upper limits) where the above circuit can be treated as an integrator. *(2 marks)*

e) Assuming an ideal opamp in the above circuit, an integrator can be realized with a transfer function Vo/Vi = Ki/s. Find the integrator gain Ki. *(1 mark)*

f) Find the difference between the gain (in dB) of the integrators in 1(d) and 1(e). *(1 mark)*

Question-2

(5 marks)

The small-signal diagram of a closed loop system is shown in Fig. 2. K = 0 unless stated otherwise. ω_{u1} = 2 Mrad/s, ω_{u2} = 200 Mrad/s, and ω_{p3} = 80 Mrad/s.

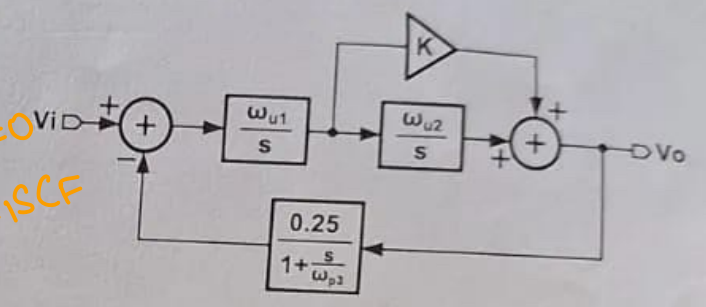


Fig. 2 Figure for Question-2.

a) Considering K=0, find the open loop gain LG(s). Find the phase margin for the above loop. *(2 marks)*

b) Gain, K is a scalar quantity and added to stabilize the loop by adding a zero in the loop. Find K such that the resulting loop has a 76° phase-margin. Find the new unity gain frequency for the stabilized loop. (Hint: Assume a wide separation between poles and zeros for the loop wherever possible.) *(3 marks)*

March 2025 9:06 am

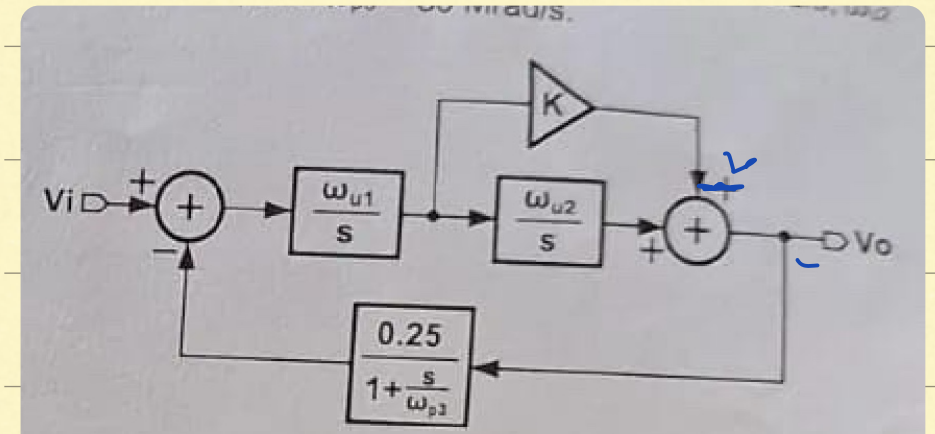


Fig. 2 Figure for Question 2

a) $\frac{\omega_1}{s}\cdot\frac{\omega_2}{s}\cdot\frac{1}{1+s/\omega_3} = \frac{2\times2\times10^2\times8\times10\times1/4}{s^2(s+8\times10)}$

$\omega^4(\omega^2+6400) = 8^2\times10^6 \Rightarrow \omega^4 = 10^4 \Rightarrow \omega = 10$

PM: $180-90-90-\tan^{-1}\left(\frac{10}{80}\right) = -7.125°$

b) $\frac{\omega_1}{s}\left(\frac{\omega_2}{s}+K\right)\left(\frac{0.25}{1+s/\omega_3}\right) = \frac{2\times10^6\times(2\times10^8+sK)\times8\times10^7}{s^2(s+8\times10^7)}$

$76 = -\tan^{-1}\left(\frac{\omega_u}{8\times10^7}\right)+\tan^{-1}\left(\frac{\omega_u}{2\times10^8/K}\right)$

$\frac{256\times10^{26}(4\times10^{16}+\omega_u^2K^2)}{\omega_u^4(\omega_u^2+16\times10^{14})} = 1$

$(|G| = -1)$

$\omega_{gc} << \omega_{p3} \Rightarrow \tan^{-1}(\omega_{gc}/80) \approx 0 \Rightarrow \omega_{ug} = 20\times10^6$ rad/s

$K = 39.4$

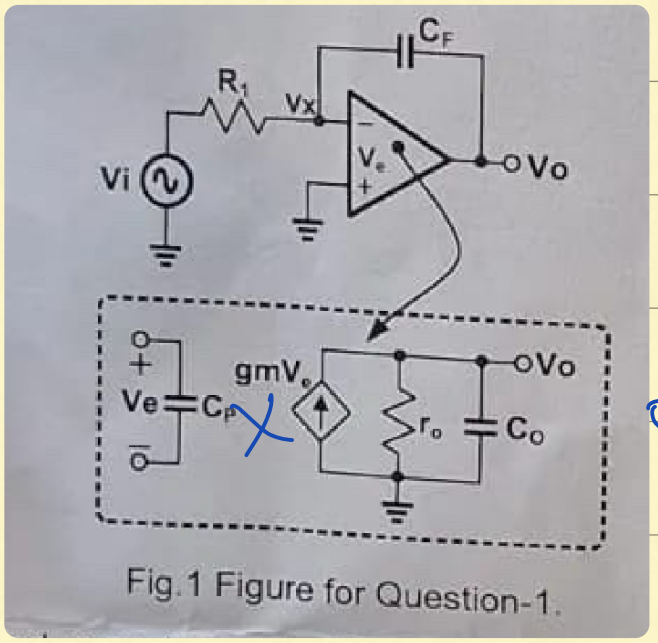
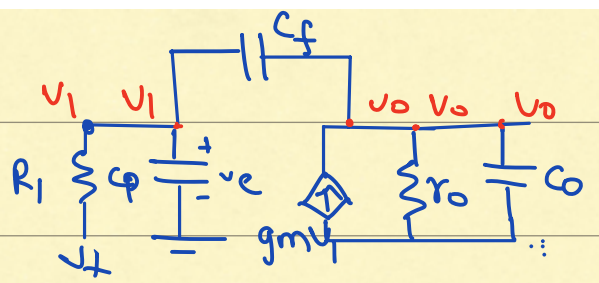


Fig.1 Figure for Question-1.

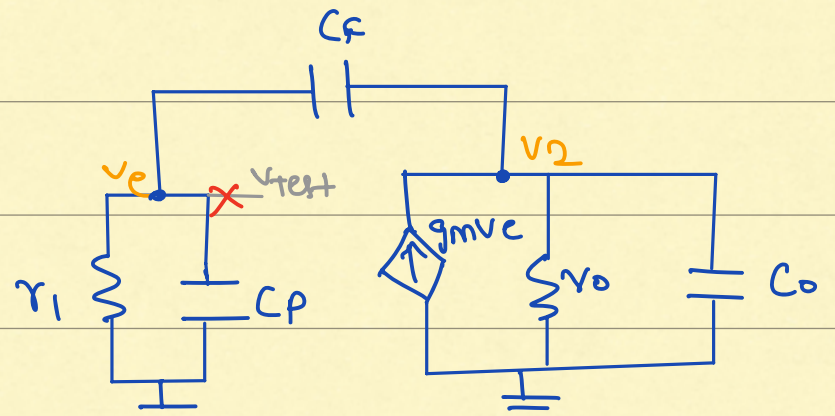


$$(V_1 - V_n)(SC_p) + V_1(G_1) + (V_1 - V_o)(SC_f) = 0$$

$$0 = -g_m V_1 + (V_o - V_1)(SC_f) + V_o(G_o) + V_o(SC_o)$$

$$\Rightarrow V_1(g_m + SC_f) = V_o(SC_f + G_o + SC_o)$$

a)

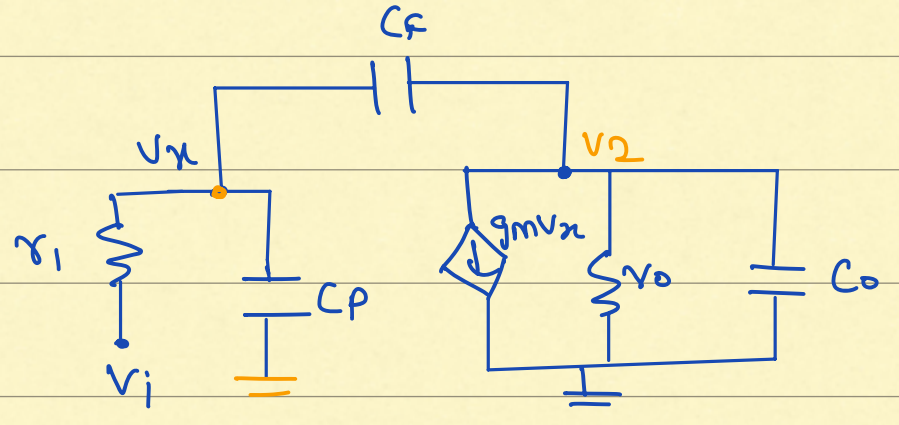


$$gmV_{test} = \frac{V_2}{r_o} + V_2(SC_o) + SC_f(V_2 - V_e)$$

$$V_e/r_1 + SC_pV_e + SC_f(V_e - V_o) = 0 \Rightarrow V_o = V_e\left[\frac{G_1 + S(C_p + C_f)}{SC_f}\right]$$

$$\frac{V_e}{V_{test}} = \frac{g_m r_o S C_f r_1}{1 + S[r_o(C_o + C_f) + r_1(C_p + C_f)] + S^2 r_o r_1 [C_p C_o + C_p C_f + C_o C_f]}$$

b)



$$(V_x - V_1)(G_1) + V_x(SC_p) + (V_x - V_2)(SC_F) = 0$$

$$V_2(G_o) + V_2(SC_o) + (V_2 - V_x)(SC_F) + g_m V_x = 0$$

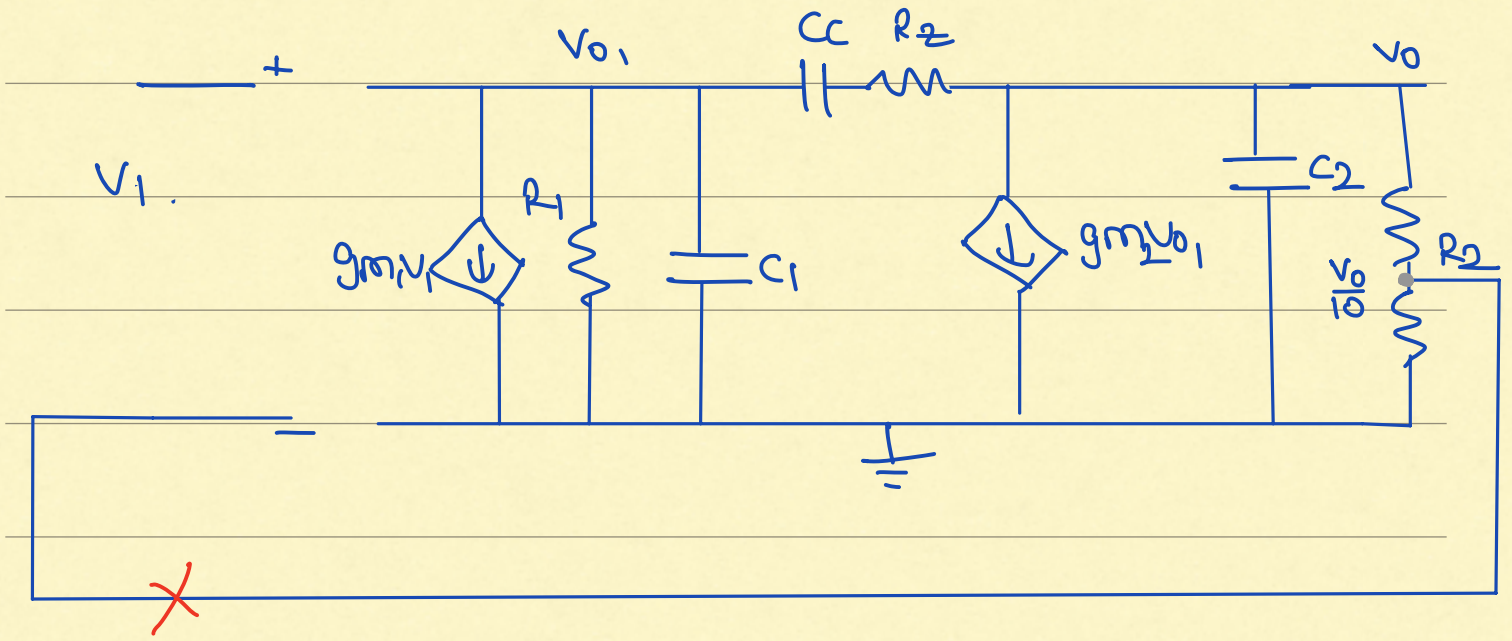
$$V_2(G_o + SC_o + SC_F) = -V_x(SC_F - g_m)$$

$$V_x(G_1 + SC_p + SC_F) - V_1(G_1) = V_2(SC_F)$$

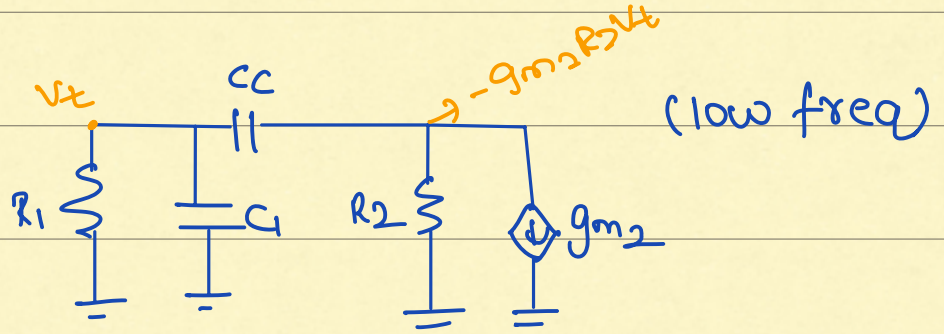
$$\frac{V_2(G_o + S(C_o + C_F))(G_1 + S(C_p + C_F))}{(SC_F - g_m)} - V_2(SC_F) = V_1(G_1)$$

$$V_2\left[\frac{S^2[(C_o + C_F)(C_p + C_F) - C_F^2] + S[G_1(C_o + C_F) + G_o(C_p + C_F) + g_m C_F] + G_o G_1}{SC_F - g_m}\right] = V_1(G_1)$$

$$\frac{V_2}{V_1} = \frac{G_1(SC_F - g_m)}{S^2[(C_o + C_F)(C_p + C_F) - C_F^2] + S[G_1(C_o + C_F) + G_o(C_p + C_F) + g_m C_F] + G_o G_1}$$



P_1 :



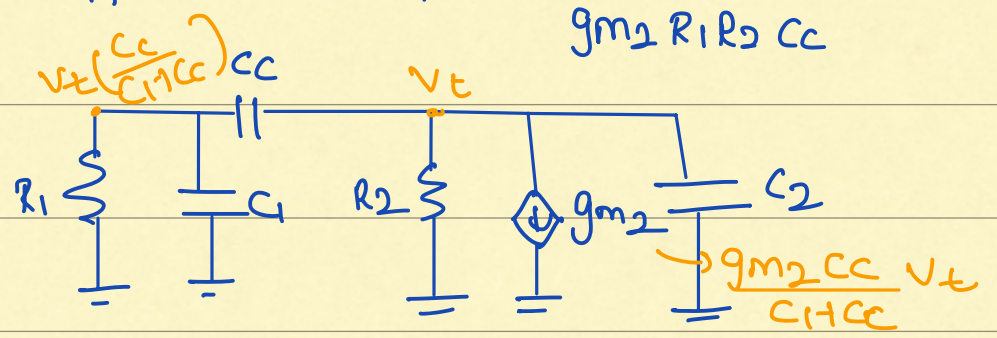
(low freq)

$\therefore$

$$C_{eff} = C_1 + C_C(1 + g_{m2}R_2) = g_{m2}R_2C_C$$

$$R_{eff} = R_1 \qquad p_1 = \frac{1}{g_{m2}R_1R_2C_C}$$

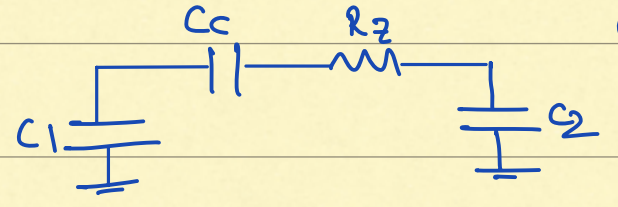
P_2 :



$$C_{eff} = C_2 + \frac{C_1C_C}{C_1+C_C}$$

$$R_{eff} = \frac{g_{m2}C_C}{C_1+C_C} + \frac{1}{R_2}$$

P_3 :



(high freq)

$$C_{eff} = \frac{1}{1/C_1 + 1/C_2 + 1/C_C}$$

$$R_{eff} = R_2$$

$$g_{m2} V_t = \frac{V_t R_2}{1 + sC_C R_2} \Rightarrow \text{zero} = \frac{g_{m2}/C_C}{1 - g_{m2} R_2}$$

$$LG = \frac{g_{m1} r_1 g_{m2} r_2}{10} \left(\frac{1 + s/z}{(1 + s/p_1)(1 + s/p_2)(1 + s/p_3)} \right)$$